

SQL & Power BI Interview Questions

Crack Interviews with Clarity



**30 Real Questions from Top Companies —
Each Verified, Solved, and Explained**

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Data Analyst Interview Questions – SQL & Power BI (30 Q&A)

1: What is the difference between INNER JOIN and LEFT JOIN?

Accenture

A:

- **INNER JOIN** returns only matching rows from both tables.
- **LEFT JOIN** returns all rows from the left table and matching rows from the right table; unmatched right rows return NULL.

2: How do you find the second-highest salary in a table?

EY

A:

Interview-Questions

SQL

```
SELECT MAX(salary) AS second_highest
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

3: What is the difference between WHERE and HAVING?

WorkIndia

A:

- **WHERE** filters rows before aggregation.
- **HAVING** filters aggregated results after GROUP BY.

4: How to find duplicate records in SQL?

A:

Capgemini

Interview-Questions

SQL

```
SELECT column_name, COUNT(*)
FROM table_name
GROUP BY column_name
HAVING COUNT(*) > 1;
```

5: Write a query to get the top 5 customers by sales.

A:

IDFC FIRST Bank

Interview-Questions

SQL

```
SELECT customer_id, SUM(sales) AS total_sales
FROM orders
GROUP BY customer_id
ORDER BY total_sales DESC
LIMIT 5;
```

6: Explain RANK() vs DENSE_RANK().

VOIS

A:

- **RANK()** leaves gaps in ranking when there are ties.
- **DENSE_RANK()** gives consecutive ranks without gaps.

7: How do you handle NULL values in SQL?

TCS

A:

Use functions like **COALESCE()** or **ISNULL()** to replace NULL with default values.

8: Write a query to calculate the running total of sales.

A:

American Express

Interview-Questions

SQL

```
SELECT order_date, sales,  
       SUM(sales) OVER (ORDER BY order_date) AS running_total  
FROM orders;
```

9: What is the difference between UNION and UNION ALL?

A:

PwC

- **UNION** removes duplicates.
- **UNION ALL** keeps duplicates.

10: How to find customers who have not placed any orders?

A:

Target

Interview-Questions

SQL

```
SELECT c.customer_id, c.customer_name  
FROM customers c  
LEFT JOIN orders o ON c.customer_id = o.customer_id  
WHERE o.customer_id IS NULL;
```

11: Explain the purpose of indexes in SQL.

A:

Steno

Indexes speed up data retrieval by reducing the number of rows scanned.

12: Write a query to get monthly sales for 2024.

A:

Interview-Questions

SQL

```
SELECT MONTH(order_date) AS month, SUM(sales) AS total_sales
FROM orders
WHERE YEAR(order_date) = 2024
GROUP BY MONTH(order_date)
ORDER BY month;
```

13: What is the difference between TRUNCATE and DELETE?

A:

Cognizant

- **DELETE** removes rows, logs each transaction, and supports a WHERE clause.
- **TRUNCATE** removes all rows without logging each row and resets the identity.

14: How to find the percentage contribution of each category to total sales?

A:

Interview-Questions

SQL

```
SELECT category,
SUM(sales) AS category_sales,
(SUM(sales) * 100.0 / SUM(SUM(sales)) OVER()) AS percentage_contribution
FROM orders
GROUP BY category;
```

15: Explain CTE and when to use it.

A:

Greystar

- **CTE (Common Table Expression)** is a temporary result set defined using WITH for better readability and reusability in complex queries.

Power BI – 15 Questions

1: What is the difference between calculated column and measure in Power BI?

Nagarro

A:

- The calculated column is stored in the model and computed row by row.
- Measure is calculated on the fly based on the filter context.

2: How do you handle data refresh in Power BI?

ClicFlyer

A:

- Schedule refresh in Power BI Service.
- Ensure the gateway is configured for on-premises sources.

3: Explain Row-Level Security (RLS) in Power BI

Genpact

A:

- **RLS** restricts data visibility based on filters defined in roles, so users only see authorised data.

4: What is the difference between Import and DirectQuery mode?

Nagarro

A:

- **Import** stores data in Power BI for faster performance.
- **DirectQuery** queries the source directly, useful for large datasets.

5: How to create year-to-date (YTD) sales in DAX?

GLT+

A:

YTD Sales =
TOTALYTD(SUM(Orders[Sales]), 'Orders'[Order Date])

6: How do you improve Power BI dashboard performance?

ClicFlyer

A:

- Reduce columns and rows in the data model.
- Use measures instead of calculated columns where possible.
- Apply aggregations and query folding.

7: Explain relationship cardinality in Power BI.

Wipro

A:

- **One-to-One** (1:1), **One-to-Many** (1:*), and **Many-to-Many** (:) define how tables relate.

8: How to create a previous year sales comparison in Power BI?

EY

A:

Previous Year Sales =
CALCULATE(SUM(Orders[Sales]),
SAMEPERIODLASTYEAR('Orders'[Order Date]))

9: What is a Star Schema in Power BI?

EY

A:

A data model design with one fact table linked to multiple dimension tables, improving query performance.

10: How do you use Bookmarks in Power BI?

EY

A:

Bookmarks capture a report's current state, allowing toggling between views for storytelling or navigation.

11: How to create a Top 5 products by sales measure in DAX?

ClicFlyer

A:

Top 5 Products Sales =
TOPN(5, SUMMARIZE(Products, Products[Product Name], "Sales",
SUM(Orders[Sales])), [Sales], DESC)

12: What is Query Folding in Power BI?

EY

A:

Query Folding pushes transformations back to the data source, reducing load time and improving performance.

13: How to create a Profit Margin % in Power BI?

A:

Profit Margin % =
DIVIDE(SUM(Orders[Profit]), SUM(Orders[Sales]), 0)

14: How do you publish and share a Power BI report securely?

EY

A:

- Publish to Power BI Service.
- Share via workspaces, apps, or RLS-based sharing.

15: How do you handle different time zones in Power BI reports?

A:

- Convert UTC to local time in Power Query using DateTimeZone.SwitchZone.